Hybrid Inheritance:

1. Write a C++ program with a base class **Person** (name, age).
   Derive **Student** (roll number, course) and **Employee** (employee id, department) from Person.
   Further derive **WorkingStudent** from both Student and Employee (working hours).
   Create an object of WorkingStudent and display all details.
2. Write a C++ program with a base class **Device** (device id, brand).
   Derive **Mobile** (RAM, storage) and **Laptop** (processor, screen size) from Device.
   Further derive **HybridDevice** from both Mobile and Laptop (supportsPenInput).
   Create an object of HybridDevice and display all details.
3. Write a C++ program with a base class **Member** (name, email).
   Derive **Teacher** (subject, experience) and **Researcher** (research area, publications).
   Derive **ResearchTeacher** from both Teacher and Researcher (projects handled).
   Create an object of ResearchTeacher and display all details.
4. Write a C++ program with a base class **Transport** (company, model).
   Derive **Car** (fuel type, seating capacity) and **Truck** (load capacity, number of wheels).
   Further derive **PickupTruck** from both Car and Truck (cargoType).
   Create an object of PickupTruck and display all details.
5. Write a C++ program with a base class **Staff** (name, id).
   Derive **Nurse** (shift, experience) and **Pharmacist** (qualification, licenseNo).
   Further derive **MedicalAssistant** from both Nurse and Pharmacist (assignedWard).
   Create an object of MedicalAssistant and display all details.

MULTIPLE INHERITANCE

1. Write a C++ program with classes **Engine** (engine number, horsepower) and **Body** (color, material).
   Derive **Car** from both Engine and Body (model name).
   Create an object of Car and display all details.
2. Write a C++ program with classes **Camera** (resolution) and **MusicPlayer** (supported formats).
   Derive **SmartPhone** from both (brand, price).
   Create an object of SmartPhone and display all details.
3. Write a C++ program with classes **EmailService** (email id) and **SMSService** (mobile no).
   Derive **Notification** (message) from both EmailService and SMSService.
   Create an object of Notification and display all details.

MULTILEVEL INHERITANCE

1. Write a C++ program with base class **SchoolMember** (name, id).
   Derive **Teacher** (subject, experience).

Further derive **SeniorTeacher** (extra duties).
Create an object of SeniorTeacher and display all details.

2. Write a C++ program with base class **Product** (name, price).
   Derive **ElectronicProduct** (warranty).
   Further derive **SmartProduct** (connectivity options).
   Create an object and display all details.
3. Write a C++ program with base class **Vehicle** (company, model).
   Derive **Car** (fuel type).
   Further derive **ElectricCar** (battery capacity, range).
   Create an object of ElectricCar and display all details.

HIERARCHICAL INHERITANCE

1. Write a C++ program with a base class Appliance (brand, power rating). Derive WashingMachine (load capacity, type – front/top) and Refrigerator (capacity in liters, has freezer or not). Create objects of both and display all details.
2. Write a C++ program with a base class Employee (emp id, emp name, basic salary). Derive FullTimeEmployee (bonus, allowances) and PartTimeEmployee (hours worked, hourly rate). Create objects of both and display their salary-related details.
3. Write a C++ program with a base class Person (name, age). Derive Student(course, marks) and Teacher (subject, salary). Create objects of both derived classes and display all details.